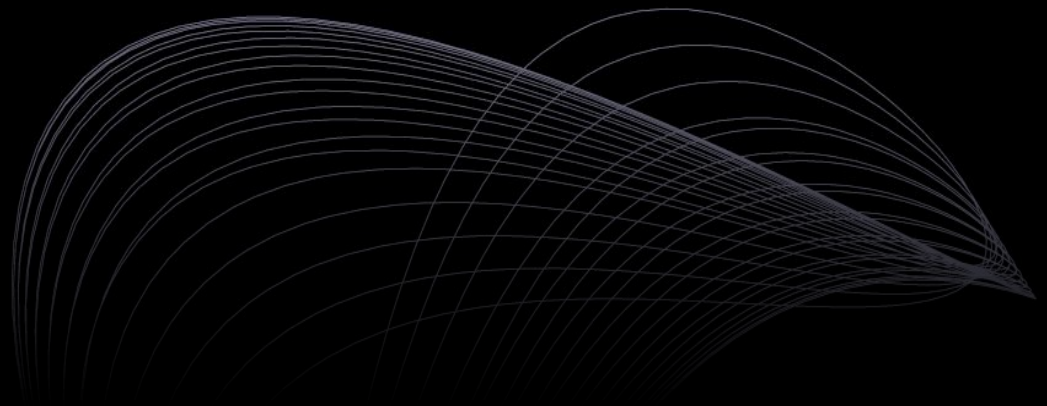
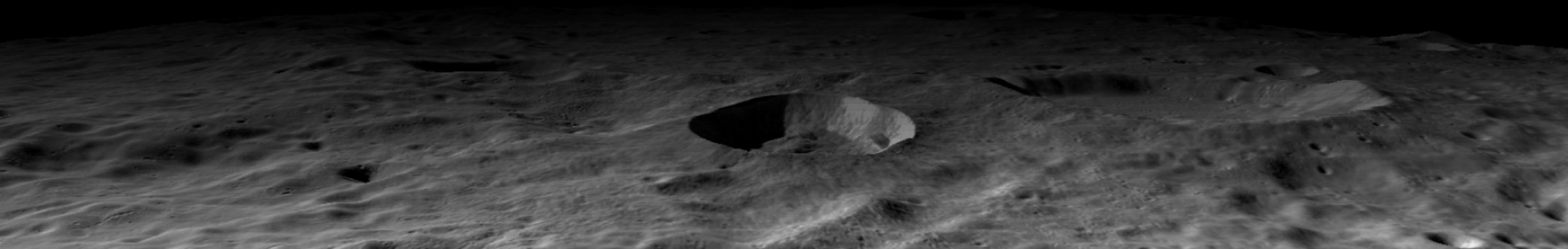
清華大學
Tsinghua University



Asteroid (469219) Kamo`oalewa’s Intriguing Journey from Lunar Crater Giordano Bruno to Earth 1:1 Resonance

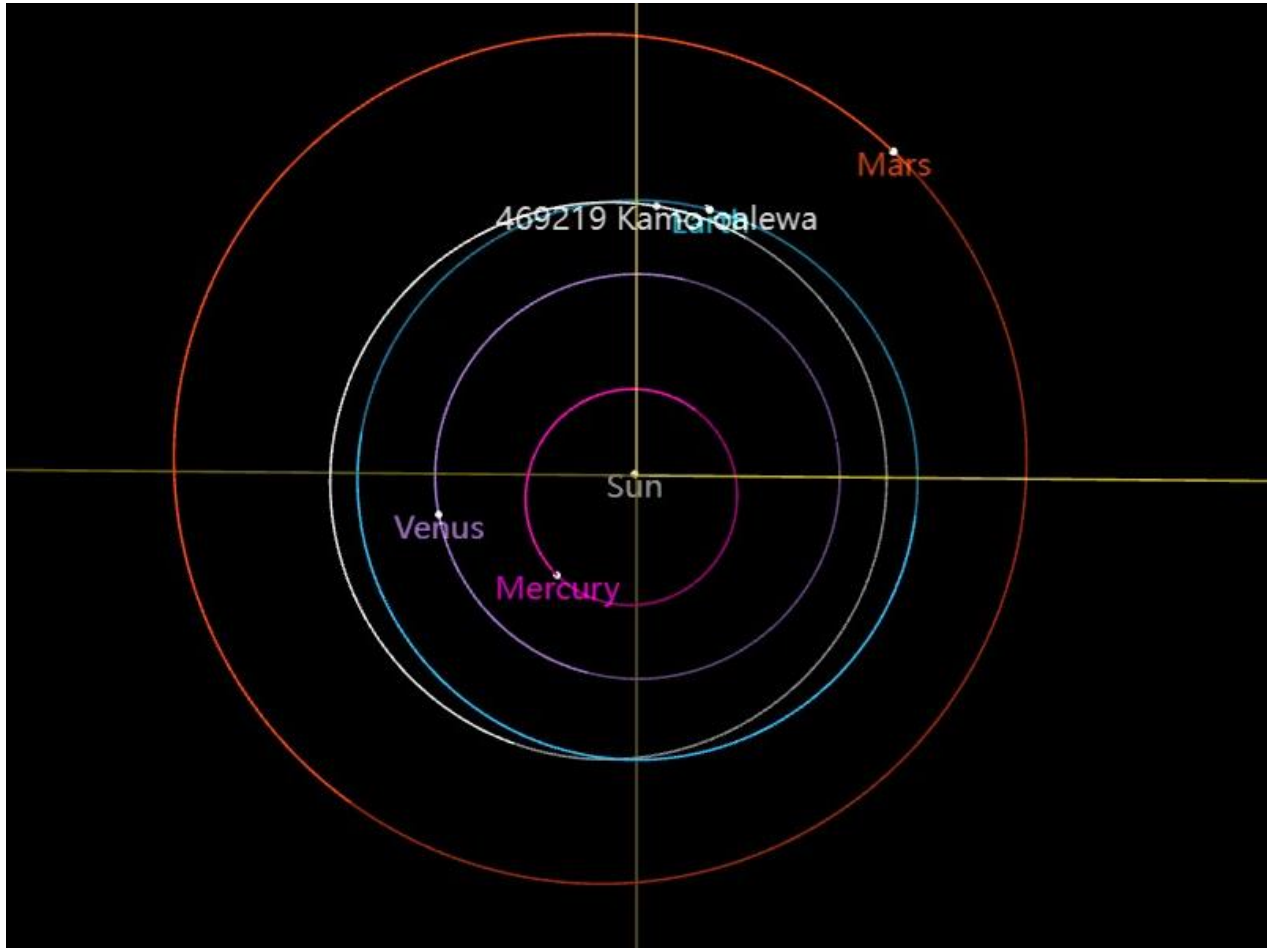


469219 Kamo`oalewa

Asteroid 469219 Kamo`oalewa (2016 HO3)

Discovery date	April 27, 2016
Orbital type	Apollo, **Quasi-Satellite**
Semi-major axis	1.001 AU
Eccentricity	0.103
Inclination	7.792 deg
Distance to Earth	38~100 LD
Velocity to Earth	2~5 km/s
Rotation period	28 min
Absolute magnitude	24.3
Mean diameter	40~60 m

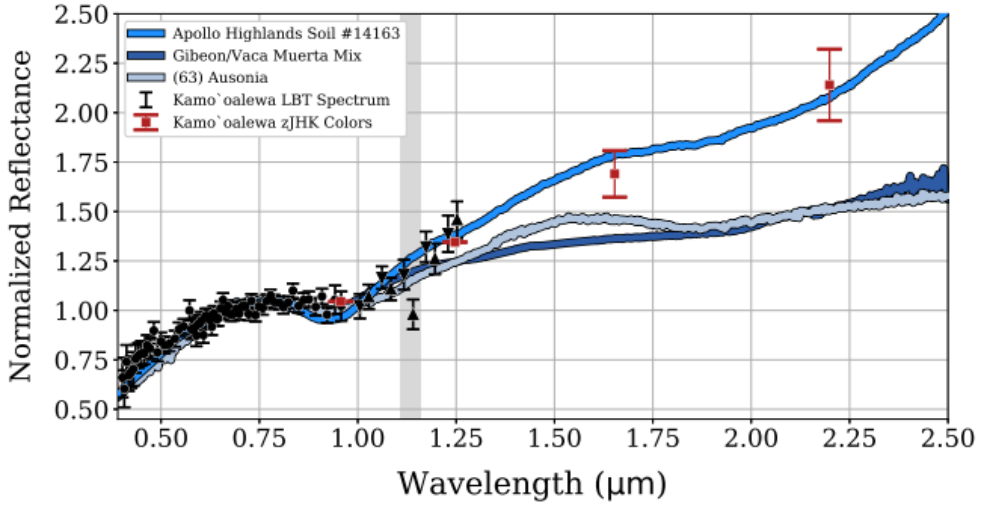
Credit: ssd.jpl.nasa.gov

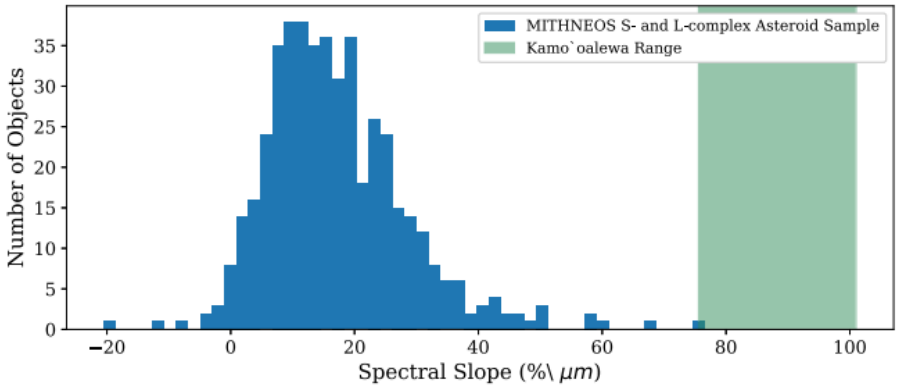


Kamo`oalewa's orbit

469219 Kamo`oalewa

Lunar-like Silicate & Earth Co-orbital Orbit

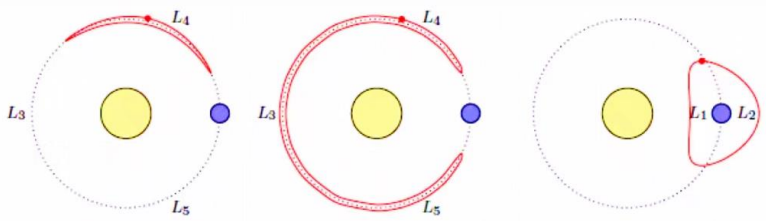


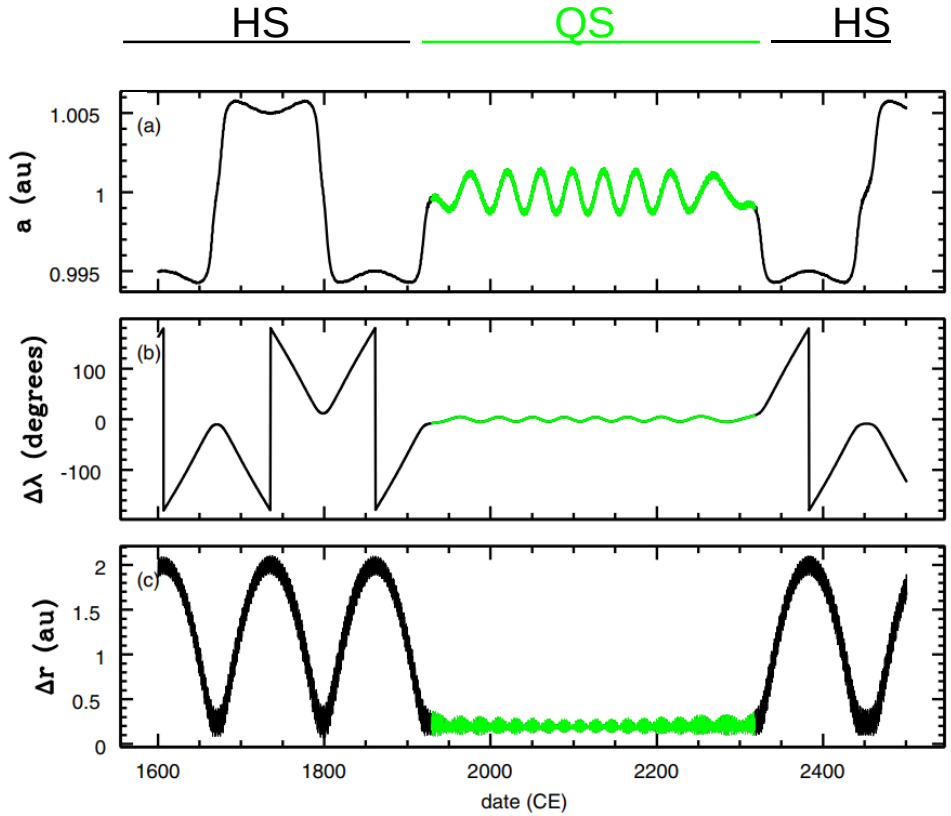


Reflectance spectra & slope

Co-orbital types:

TP, HS, QS



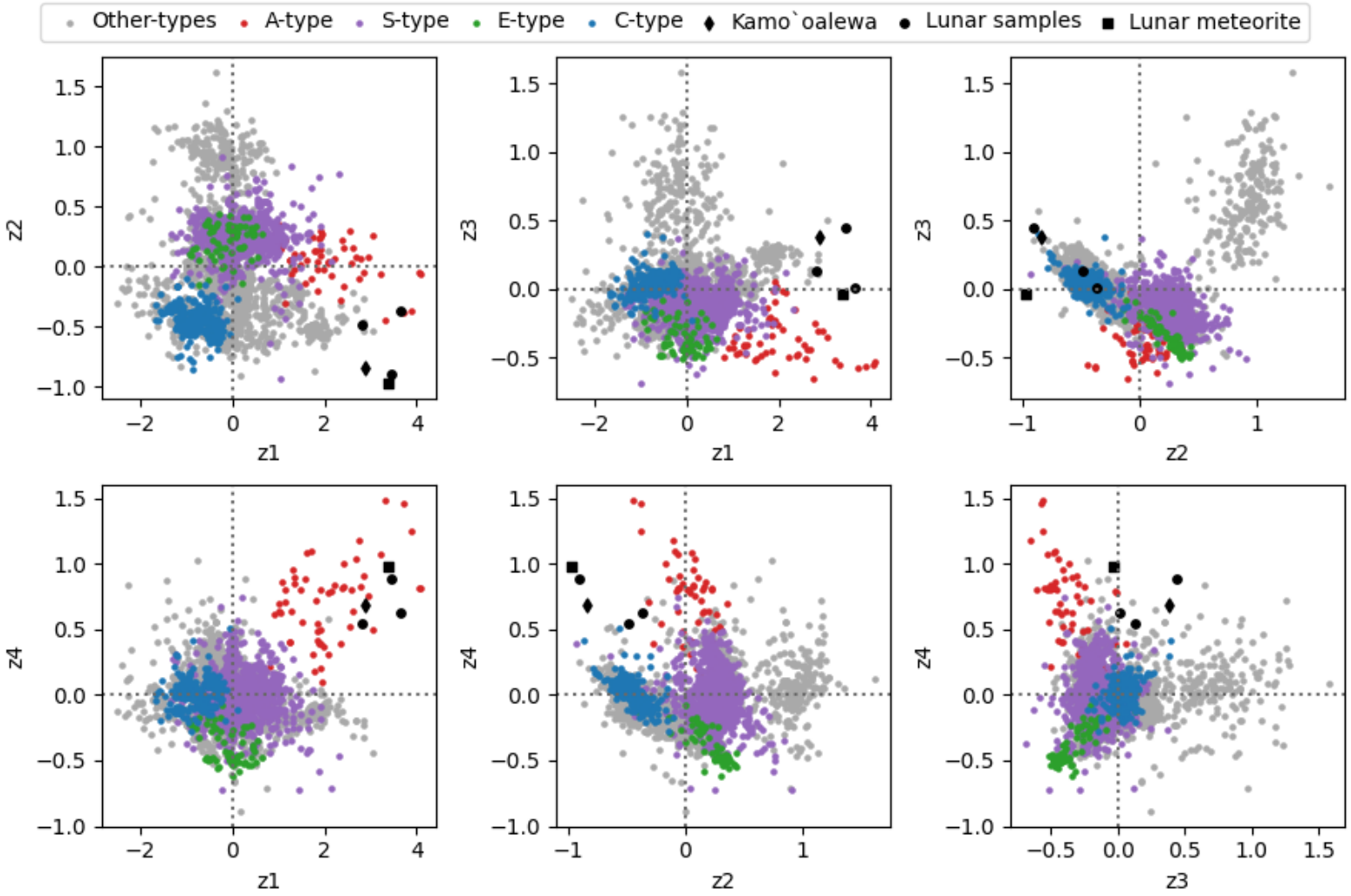


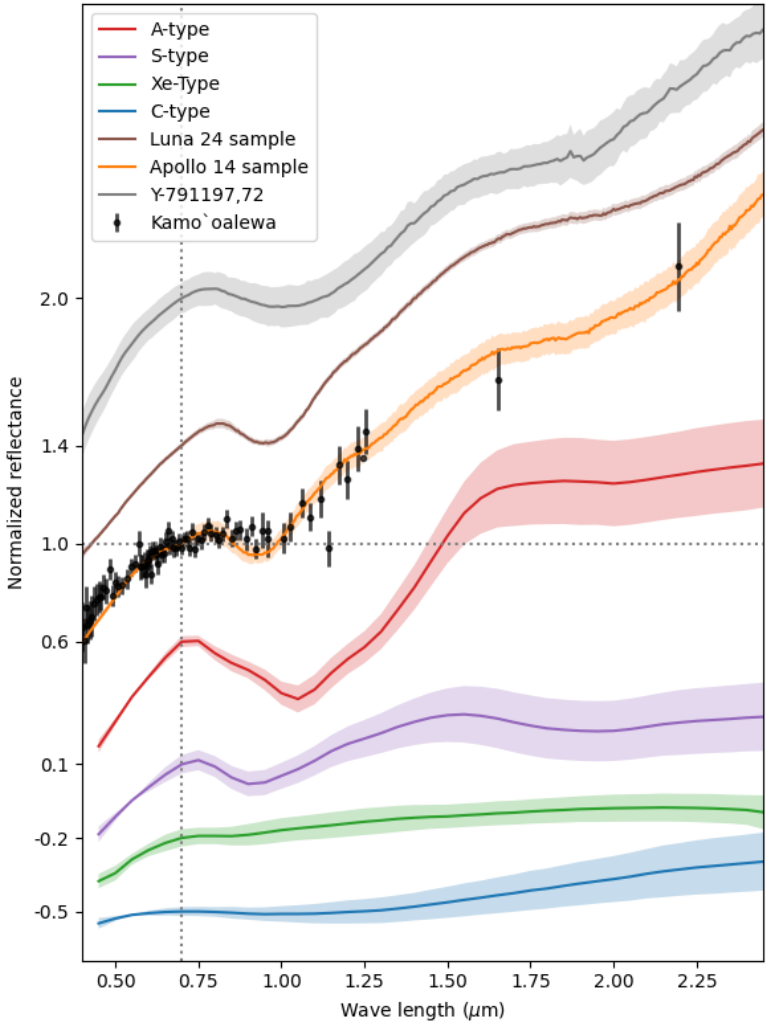
Earth co-orbital state

Credit: B. N. L. Sharkey et al. Communications Earth & Environment (2021)

Reflectance Spectra

Cluster Analysis





Credit: [1] B. N. L. Sharkey et al. Communications Earth & Environment (2021); [2] F. E. DeMeo et al. Icarus (2009); [3] sites.brown.edu/relab/relab-spectral-database; [4] space-classy by M. Mahlke et al. (2022)

A Lunar Ejecta Origin?

Kamo`oalewa's Journey

From Lunar Ejecta to NEA

Mean impact velocity[1]

Near-Earth asteroids	**18 km/s**
Jupiter-family comets	25 km/s
Parabolic comets	55 km/s
Mean impact angle	**45 deg**

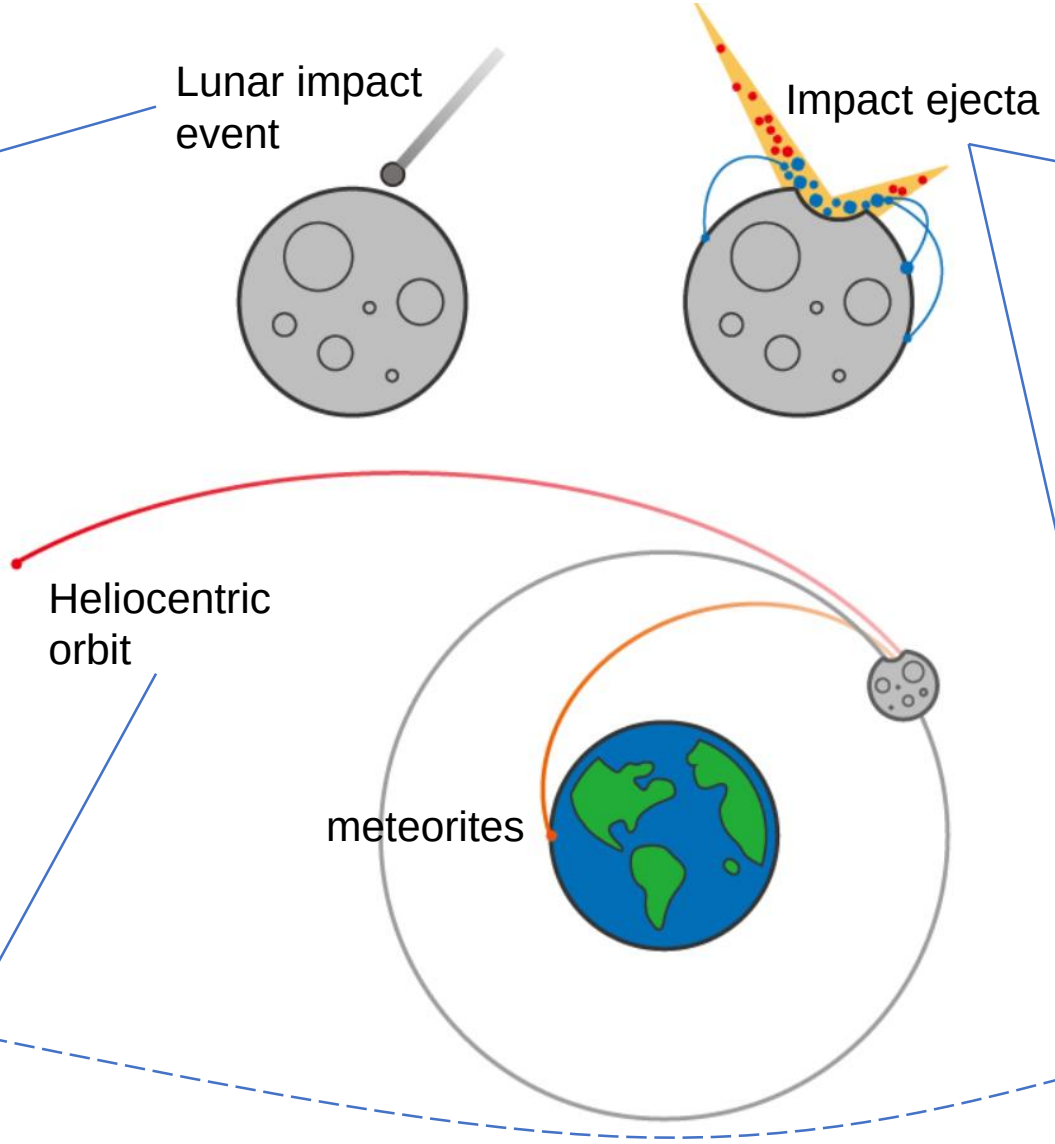
Earth impact flux from comets is below 1% than from NEAs[2]

NEAs dynamical lifetime[4,5]

Mean lifetime ~10 Myr

0.4~37.2 Myr from different ERs

10~100 Myr for collision disruptions



Low-speed ejecta

Accretion on Moon: rays

Secondary craters

Lunar Escape 2.38 km/s

High-speed ejecta

Collisions with terrestrial planets or Moon

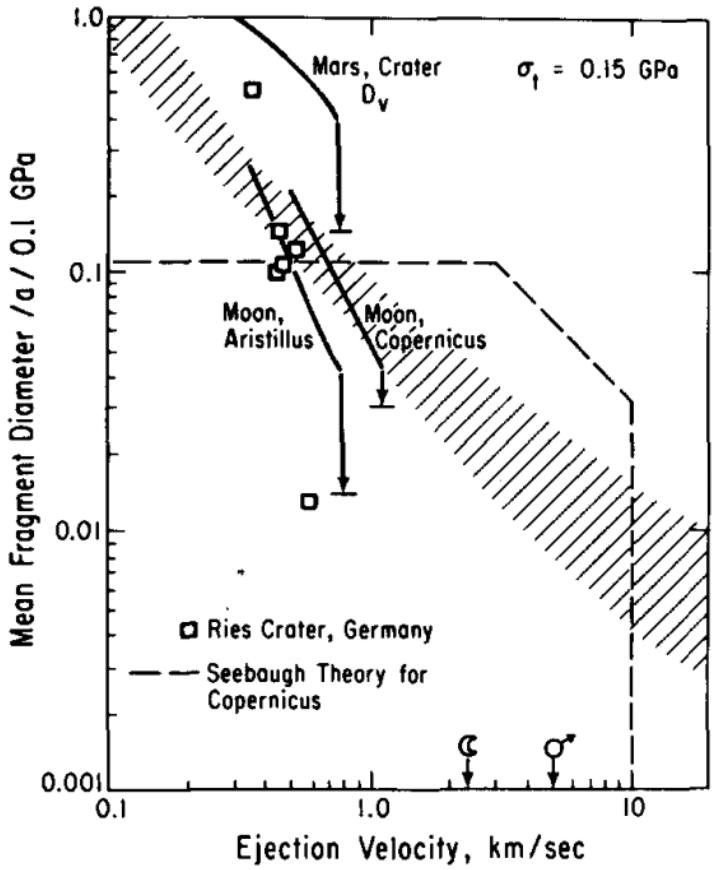
Entering heliocentric orbits[3]

Credit: [1] N. A. Artemieva et al. Solar system research (2008); [2] D. K. Yeomans et al. Acta Astronautica (2013); [3] B. J. Gladman et al. Icarus (1995); [4] M. Granvik et al. Icarus (2018); [5] W. F. Bottke et al. LPSC (1993)

Fragments Size-Velocity Power Law

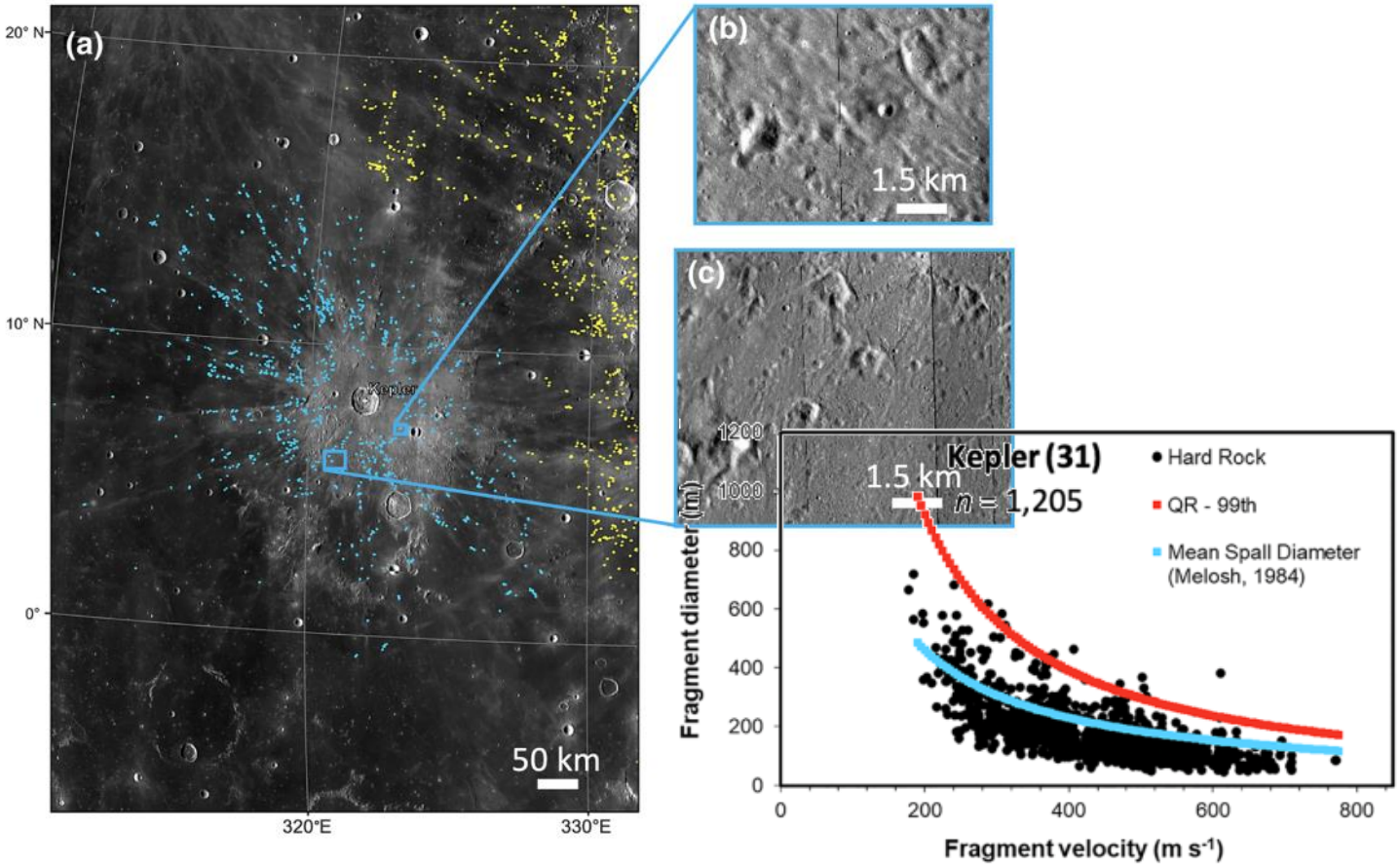
$$D_{frag}(>v) = \alpha v^{-\beta}$$

Theory



Mean fragments size-velocity[1]

Observation



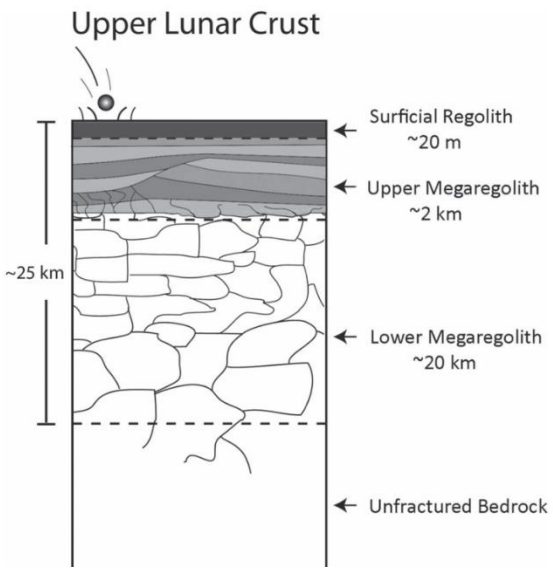
Kepler's secondary craters[2]

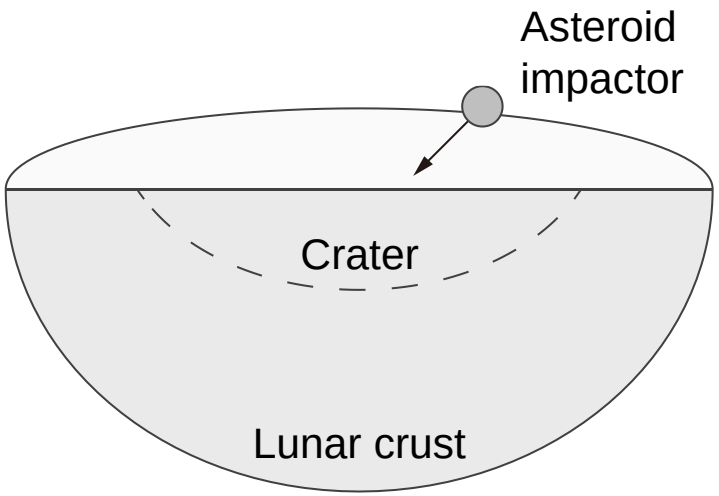
Credit: [1] H. J. Melosh, Icarus (1984); [2] K. N. Singer et al, JGR Planets (2020)

Lunar Impact Simulation

Numerical Simulations of Hypervelocity Impact

- Method: self-developed hydrocode **SPHSOL**[1]
- Case: 2-km asteroid impacts lunar crust at 18 km/s, 45 deg; total time 60 s; using 3.5 million particles (resolution 111-m), compared with higher resolution 67-m (17 million particles)





Lunar crust / asteroid impactor	
Bulk density	2700 kg/m^3
Equation of state	Tillotson (basalt)
Frictional coefficient (damaged)	0.71
Frictional coefficient (intact)	1.1
Cohesive strength (damaged)	10 kPa
Cohesive strength (intact)	31.9 MPa
Strength limit	2.49 GPa
Weibull k	10^{23} m^{-3}
Weibull m	9.5
Estimated tensile strength	10 MPa

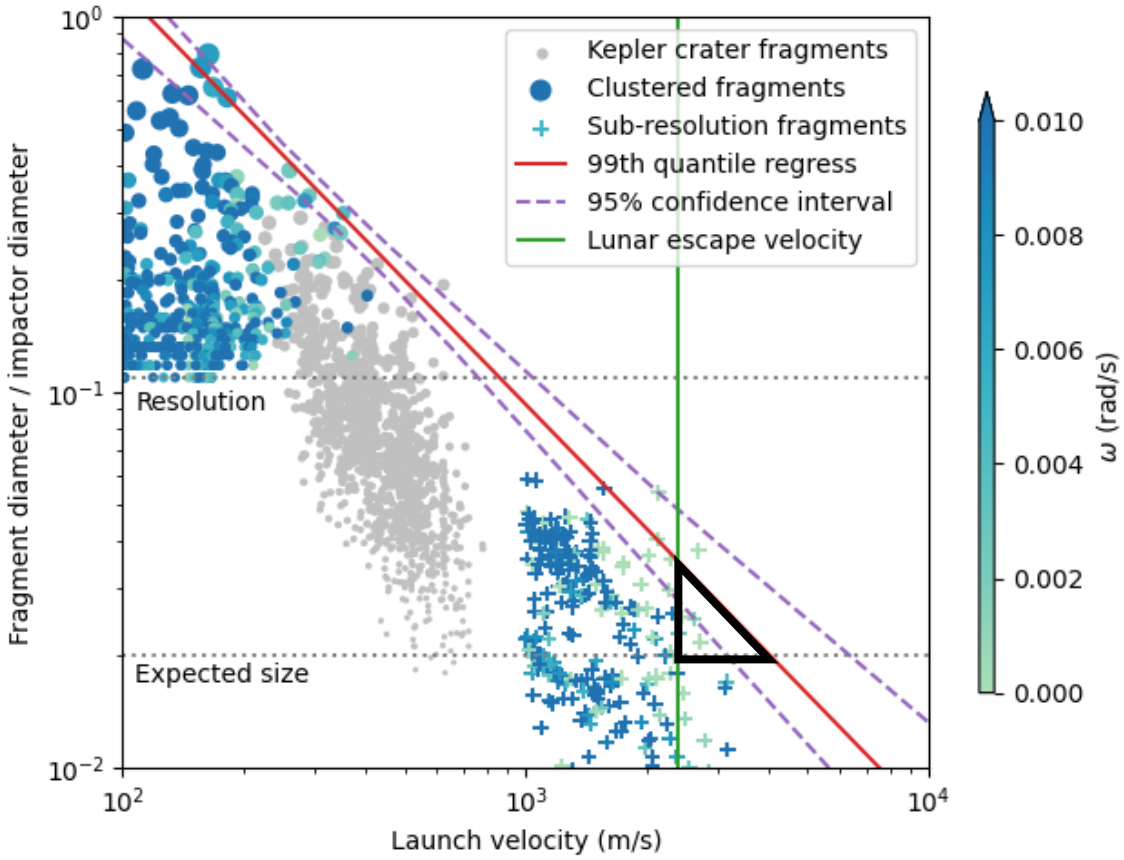
Material parameters[2,3]

Credit: [1] Yifei Jiao et al. SCIENTIA SINICA Technologica (2022); [2] R. W. K. Potter et al. Icarus (2012)
[3] Wiggins S E et al. JGR Planets (2019)

Lunar Impact Simulation

Fragment Size Upper Limit

$$D_{frag,\max}(>v) = \alpha v^{-\beta}$$



Fragments size-velocity distribution

Kepler crater (31 km): estimated from secondary craters[1]

$$\ln\alpha = 13.39 \pm 1.10, -\beta = -1.24 \pm 0.19$$

$$\ln\alpha = 12.92 \pm 1.20, -\beta = -1.17 \pm 0.20$$

Our simulated fragments

$$\ln\alpha = 12.834 \pm 0.369, -\beta = -1.102 \pm 0.060$$

$$\left(\frac{D_{\max,frag}}{D_{imp}}\right)_{escape} = 0.0356^{+0.0131}_{-0.0077}$$

To obtain at least 40-m escaped fragments:

- Impactor size > 0.8~1.4 km
- **Crater size > 10 km**

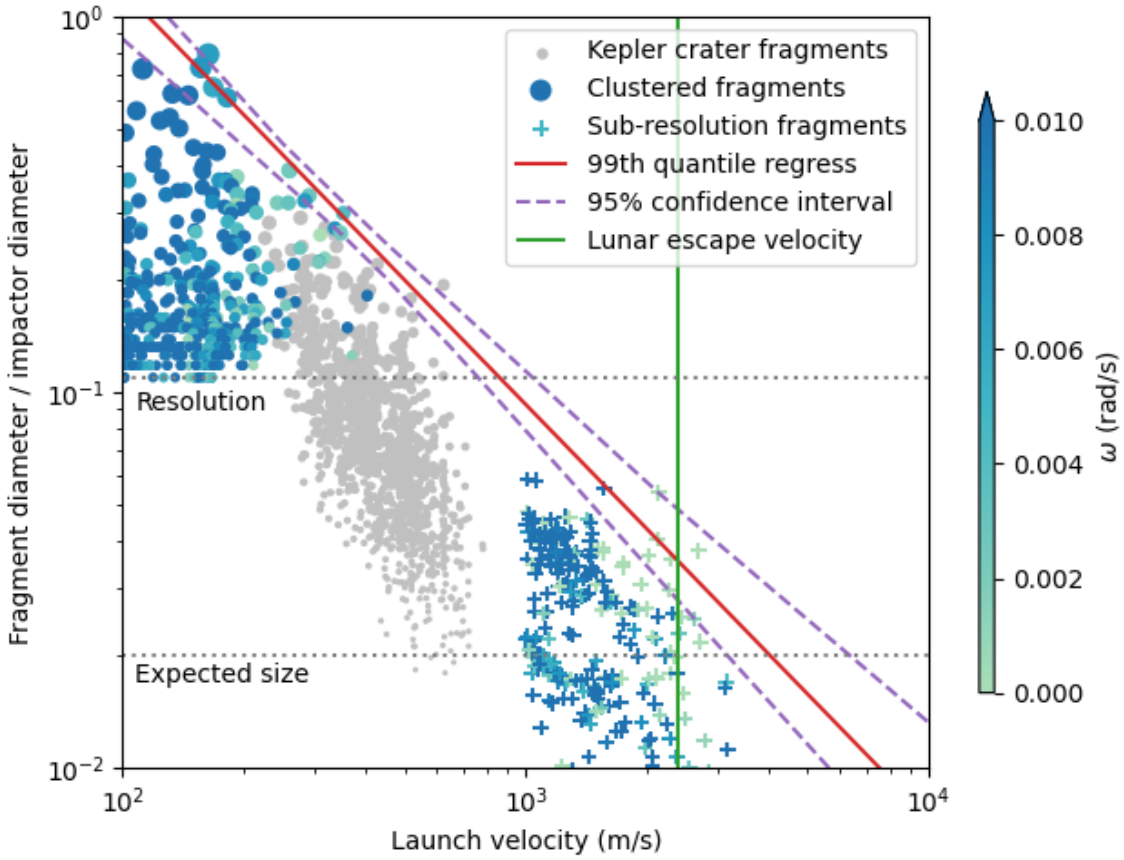
To keep NEA state until now:

- **Crater age ~ 10 Myr**

Credit: [1] K. N. Singer et al, JGR Planets (2020)

Comparison of Different Resolutions

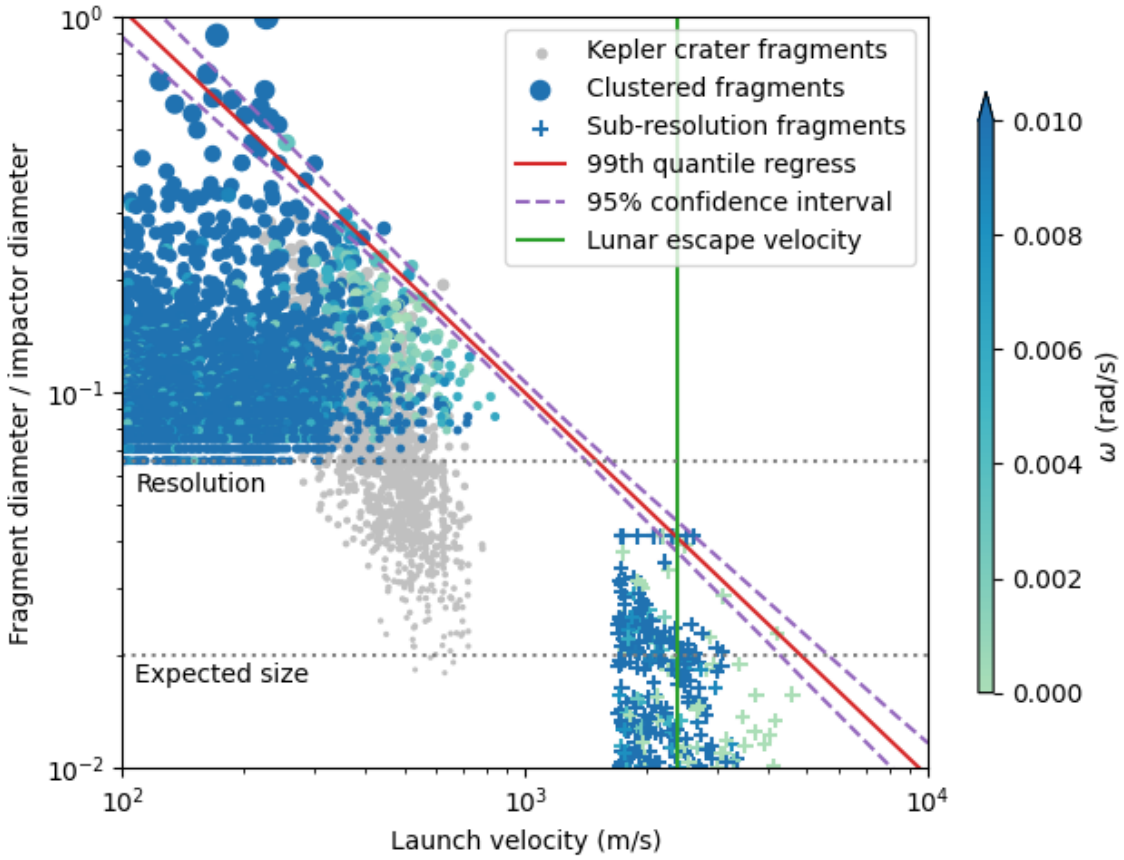
$$\ln\alpha = 12.834 \pm 0.369, -\beta = -1.102 \pm 0.060$$



Resolution 111-m

(3.5 million particles)

$$\ln\alpha = 12.338 \pm 0.205, -\beta = -1.0199 \pm 0.034$$



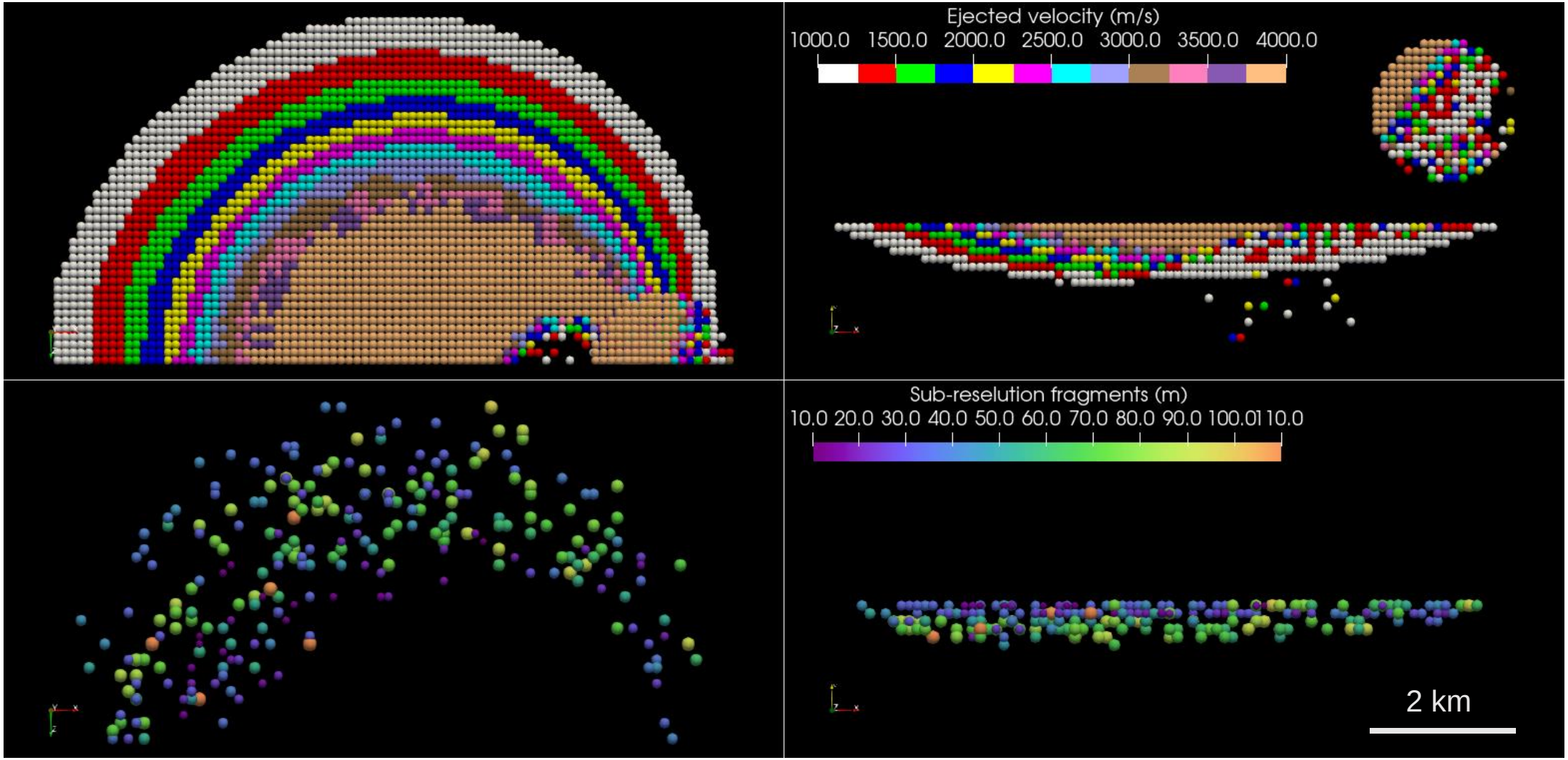
Resolution 67-m

(17 million particles)

Lunar Impact Simulation

High-Speed Impact Ejecta

Total escaping mass ~2.5 impactor mass

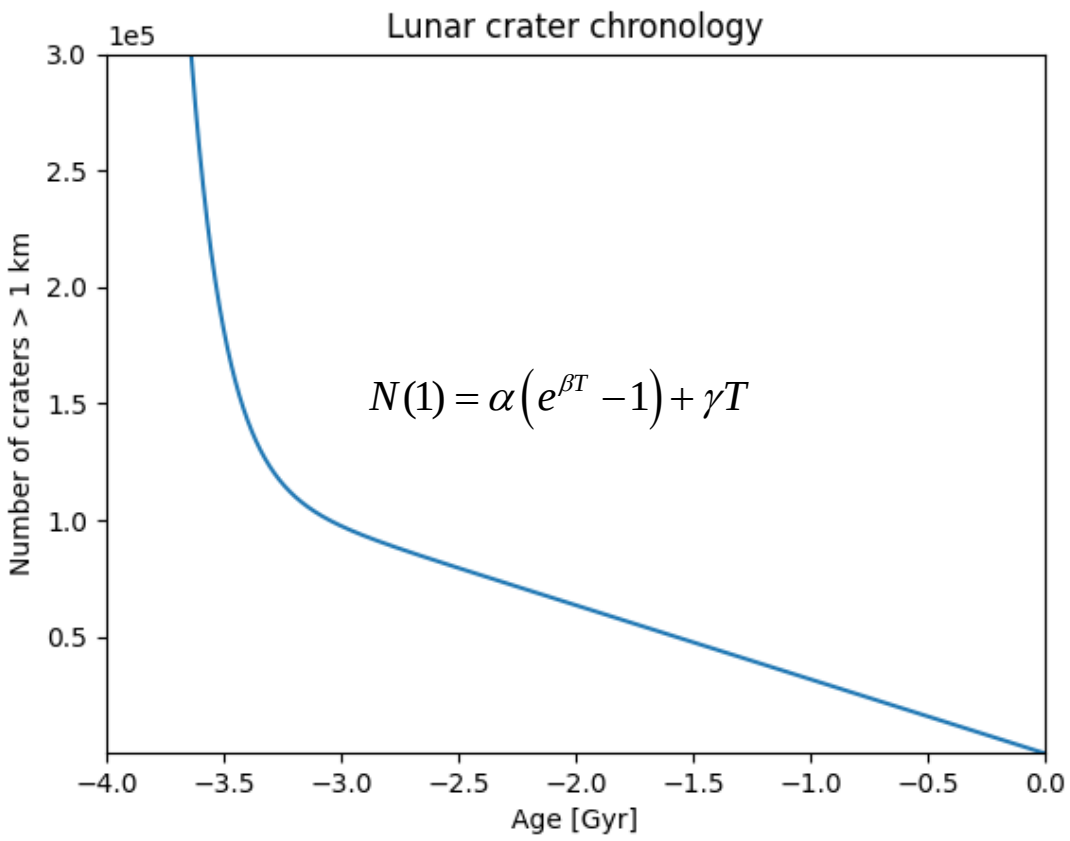


Initial position of particles that will be ejected > 1 km/s

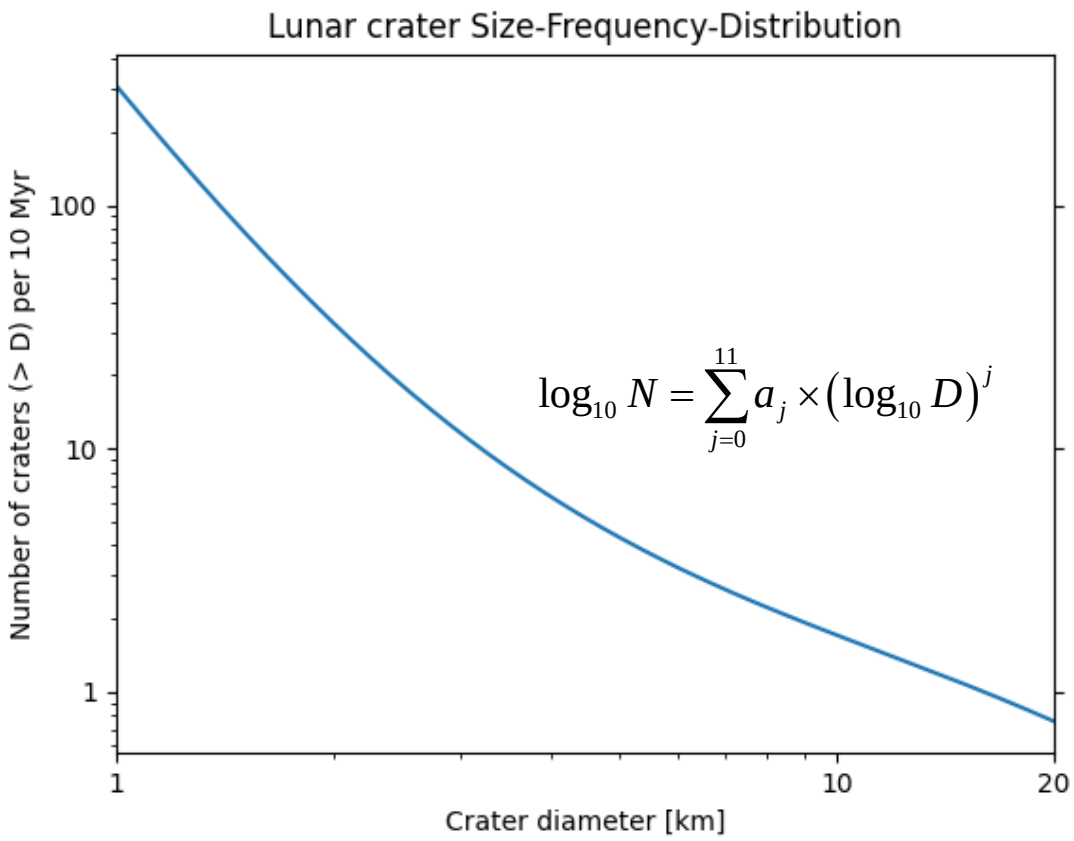
Lunar Craters

Lunar crater chronology & Distribution

- ~300 craters larger than 1-km in 10 Myr
- **Very few craters larger than 10-km in 10 Myr**



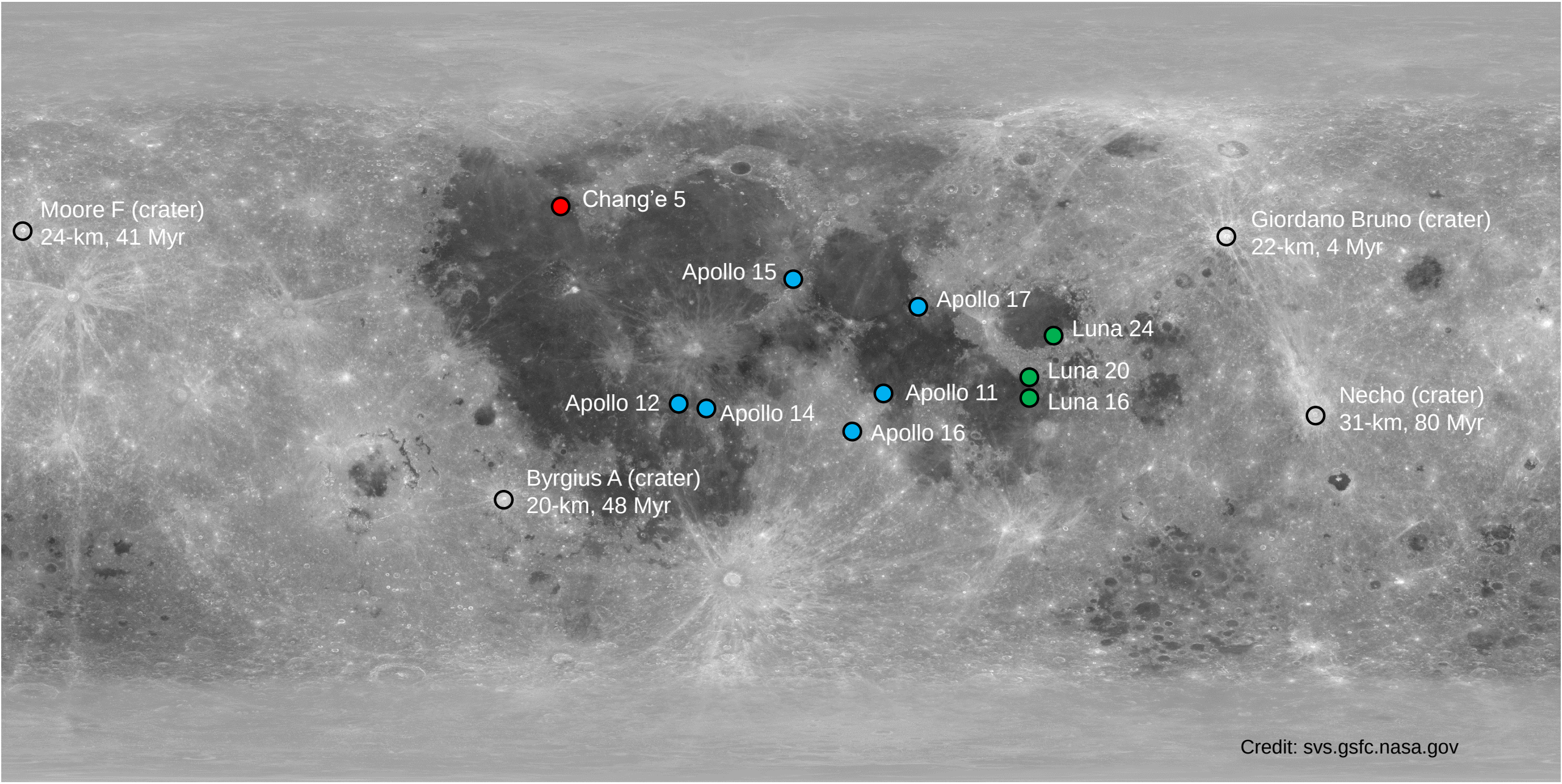
Constant production rate in 3 Gyr



Approximately power-law crater SFD

Credit: G. Neukum et al. Space Science Reviews (2001)

Lunar Map



Credit: svs.gsfc.nasa.gov

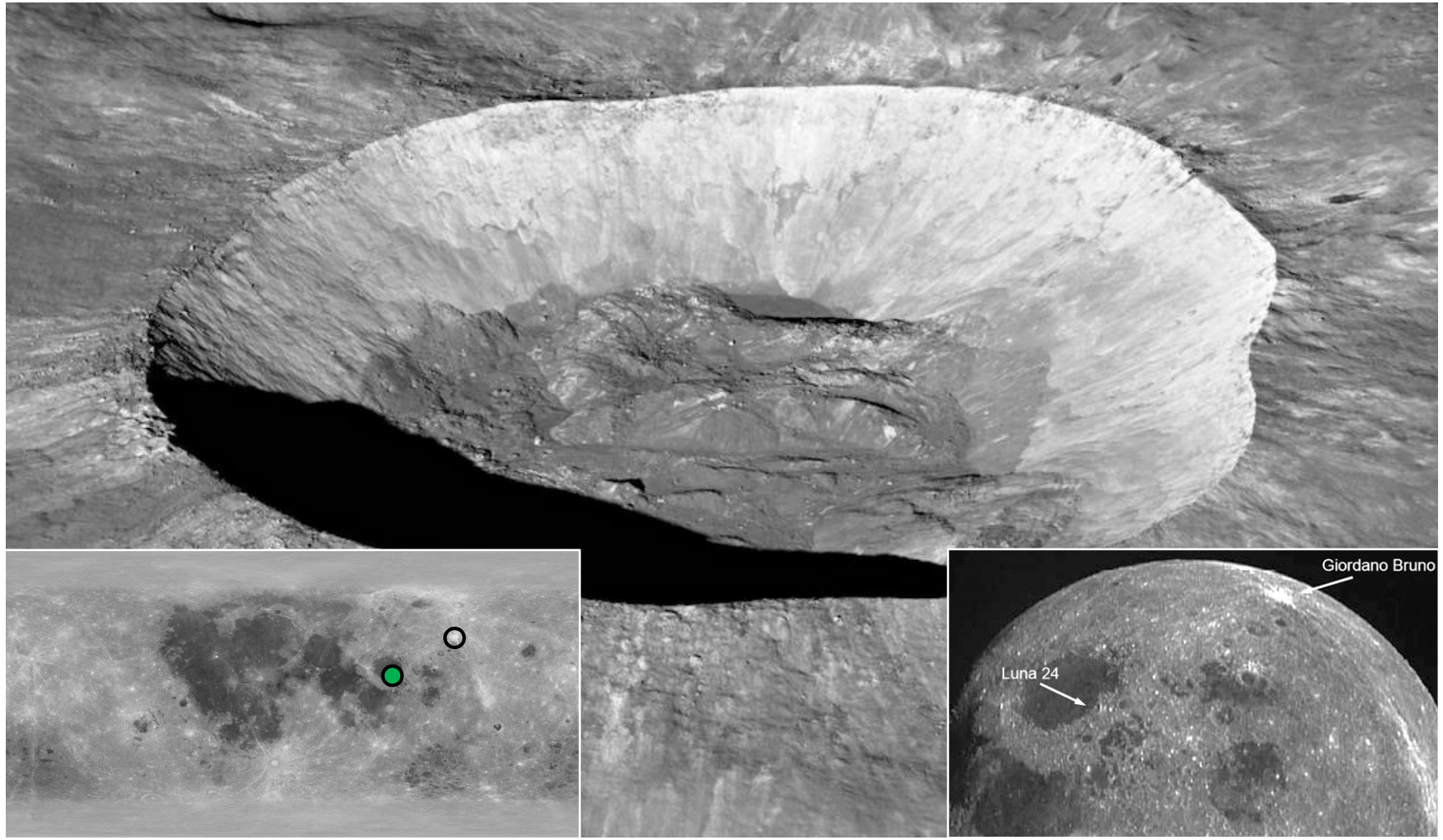
Kamo`oalewa's Source Craters

Giordano Bruno Crater

Location	Eastern & Far side
Coordinate	35.9°N 102.8°E
Diameter	22 km
Age	1~10 Myr

Most possible

- **Crater size > 10 km**
- **Crater age ~ 10 Myr**



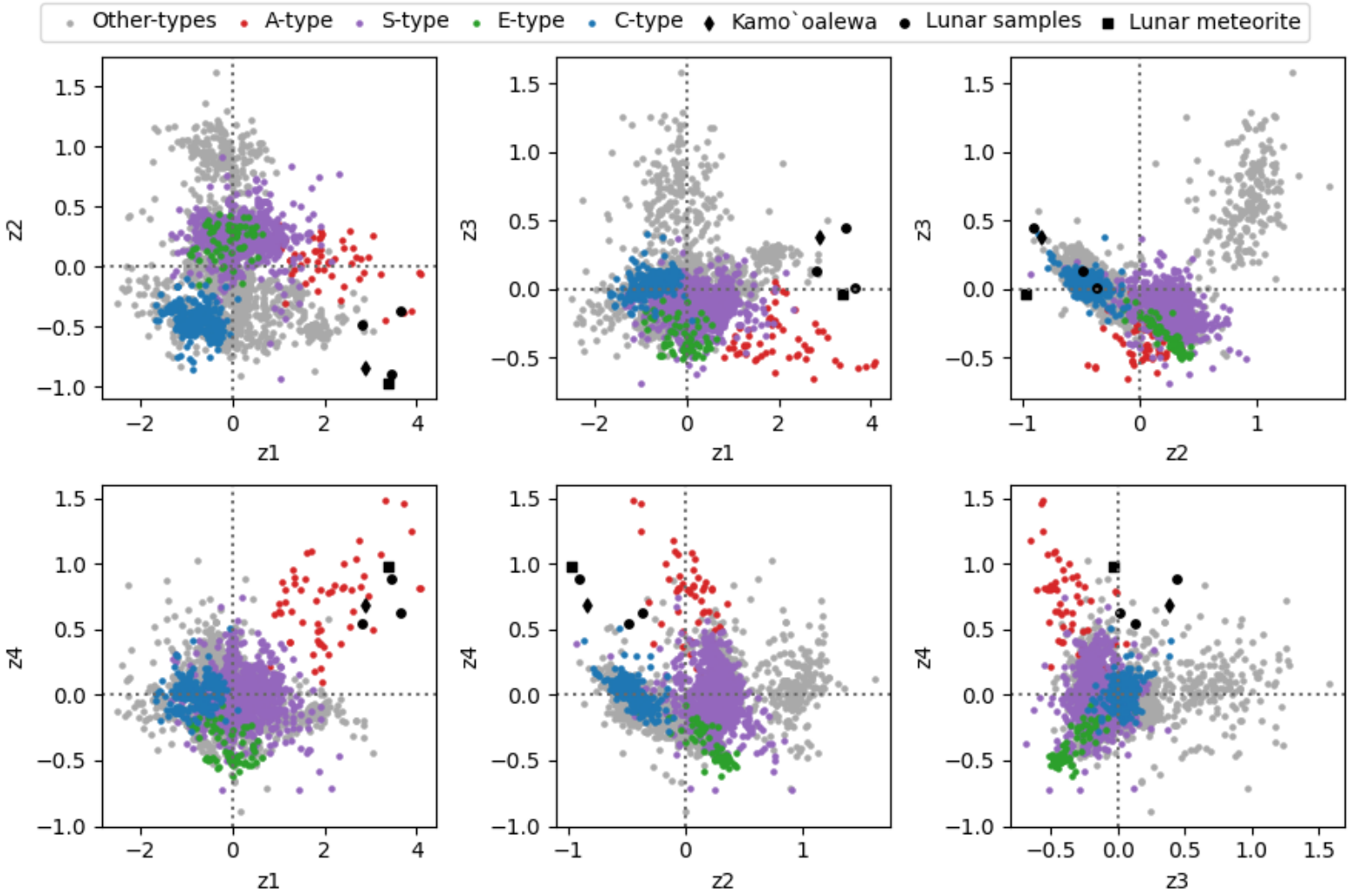
GB crater & Luna 24

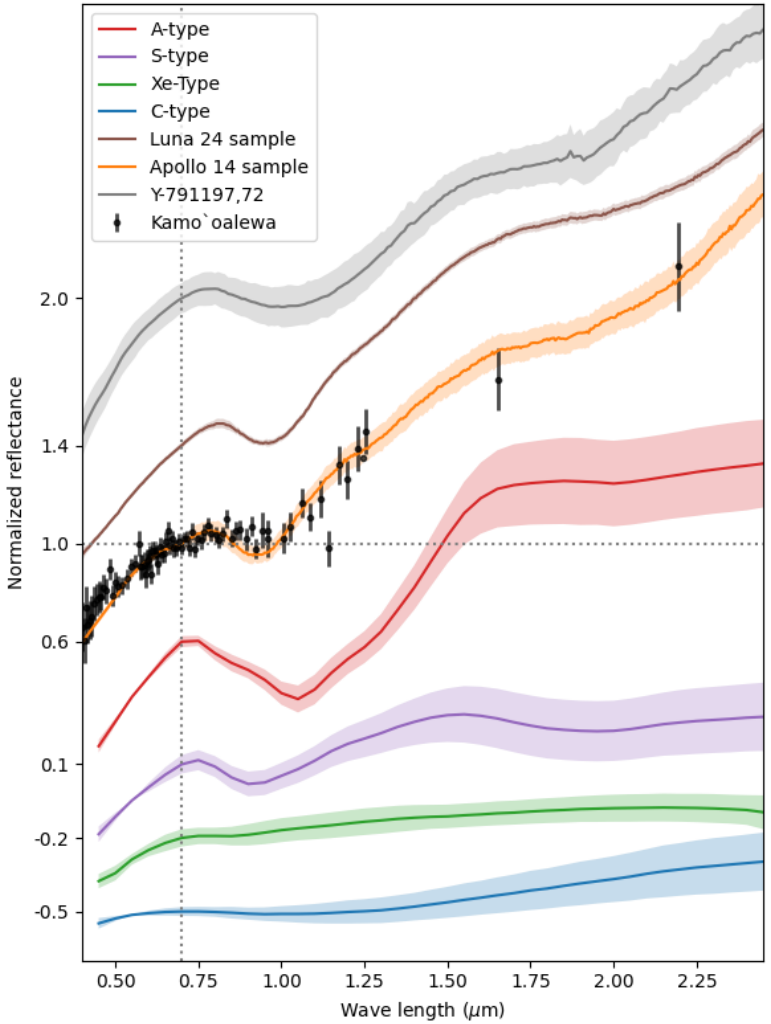
Credit: [1] lroc.sese.asu.edu; [2] A. T. Basilevsky et al. PLANET SPACE SCI (2012)

Reflectance Spectra

Cluster Analysis

Luna 24 samples may contain materials ejected from GB crater!





Credit: [1] B. N. L. Sharkey et al. Communications Earth & Environment (2021); [2] F. E. DeMeo et al. Icarus (2009); [3] sites.brown.edu/relab/relab-spectral-database; [4] space-classy by M. Mahlke et al. (2022)

Lunar Impact Simulation

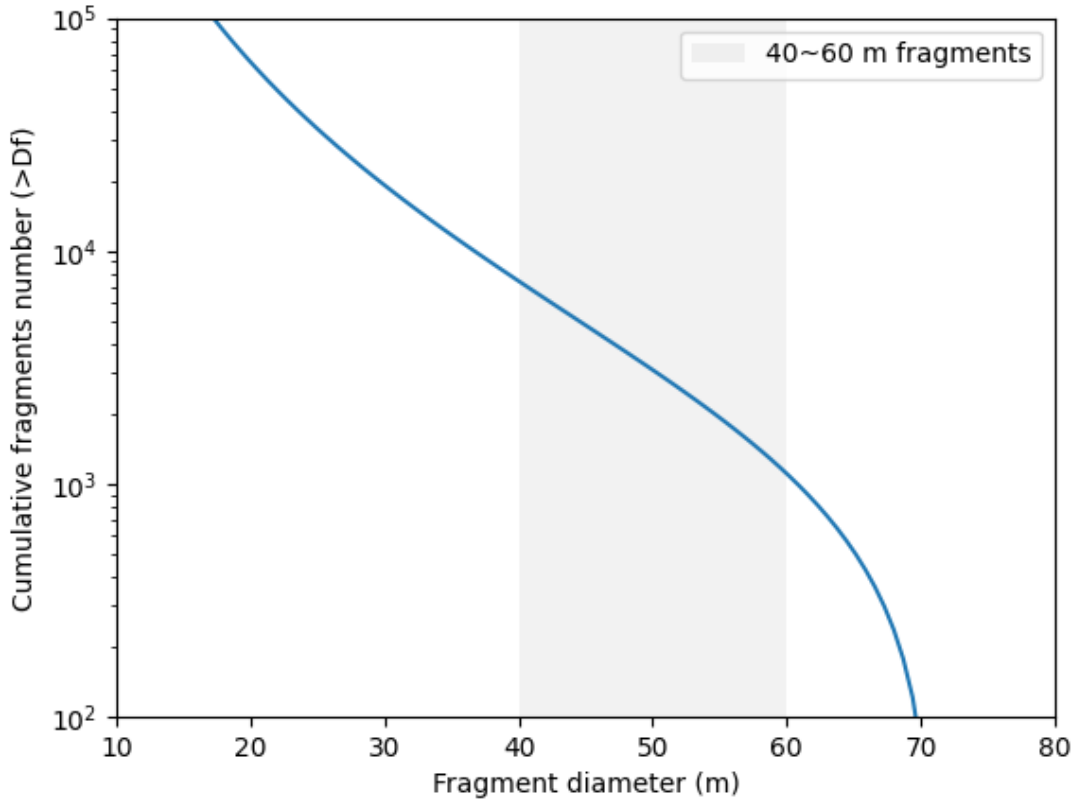
Fragments SFD & Rotation

$$N\left(D_1 < D < D_2\right) = N_0\left(D_1^{-b} - D_2^{-b}\right)$$

< 71 m

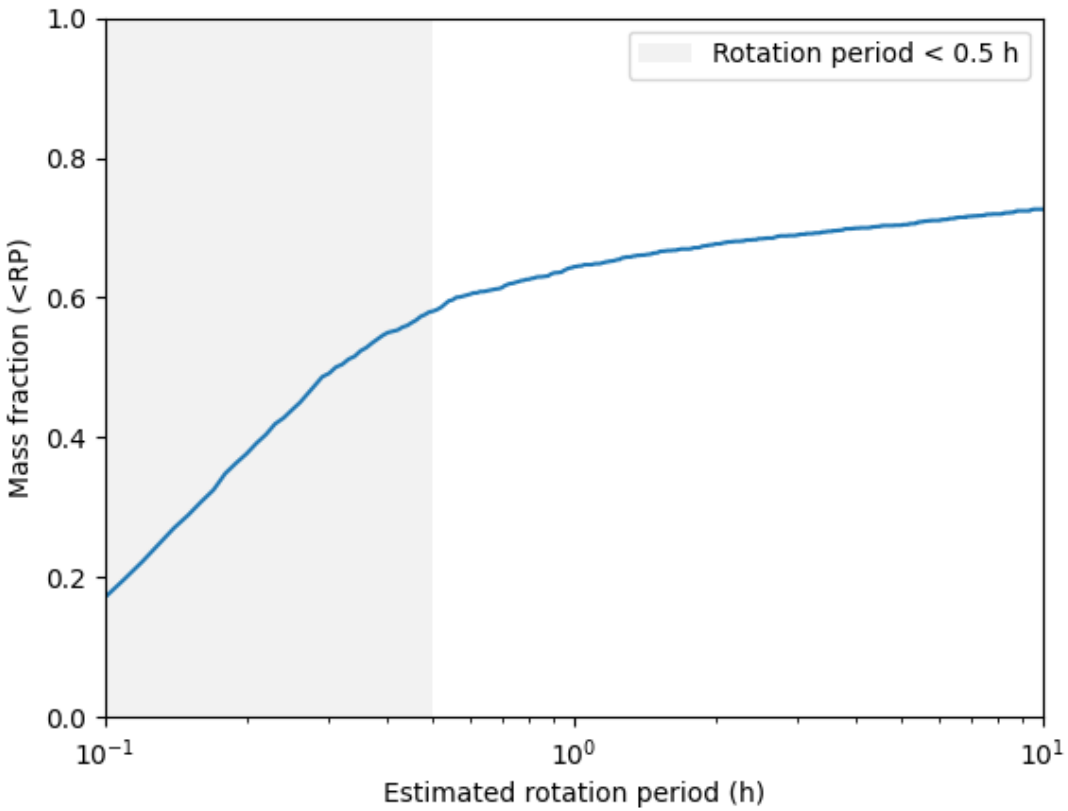
$$M\left(v_1 < v < v_2\right) = \int_{D_{\min}}^{D_{\max}} \left[\rho \frac{4\pi}{3}\left(\frac{D}{2}\right)^3 N_0 b D^{-b-1}\right] dD$$

2.38~4 km/s



Escaped Fragments SFD

~6,300 escaped fragments in 40~60 m



Escaped fragments rotation

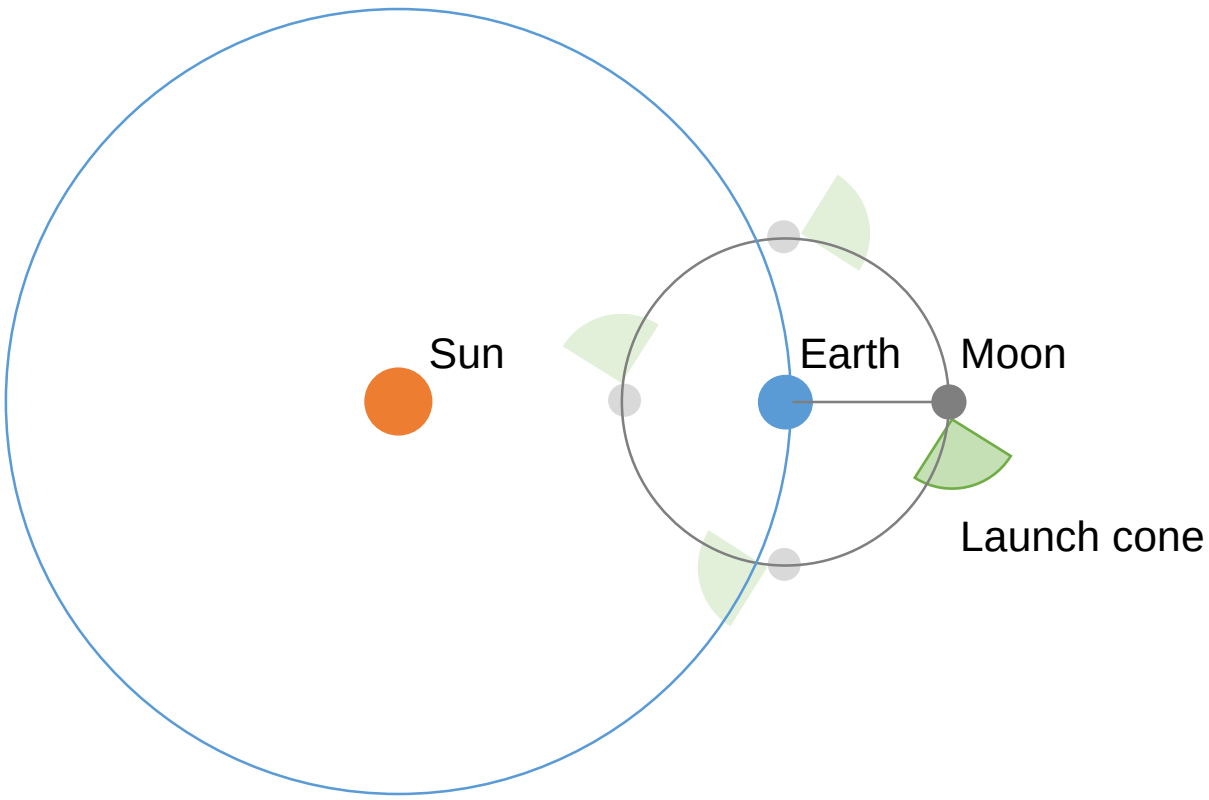
~58% rotation period < 0.5 h

Fragments Dynamical Evolution

N-body Simulations

~6,300 escaped fragments in 40~60 m at 2.38~4 km/s

- Method: open-source N-body code **REBOUND**[1]
- Case: test particles launch from GB crater (35.9°N, 102.8°E) at 2.4~3.5 km/s, 45 deg; considering different azimuths (0~360 deg) and Moon phases (0~27.3 d)



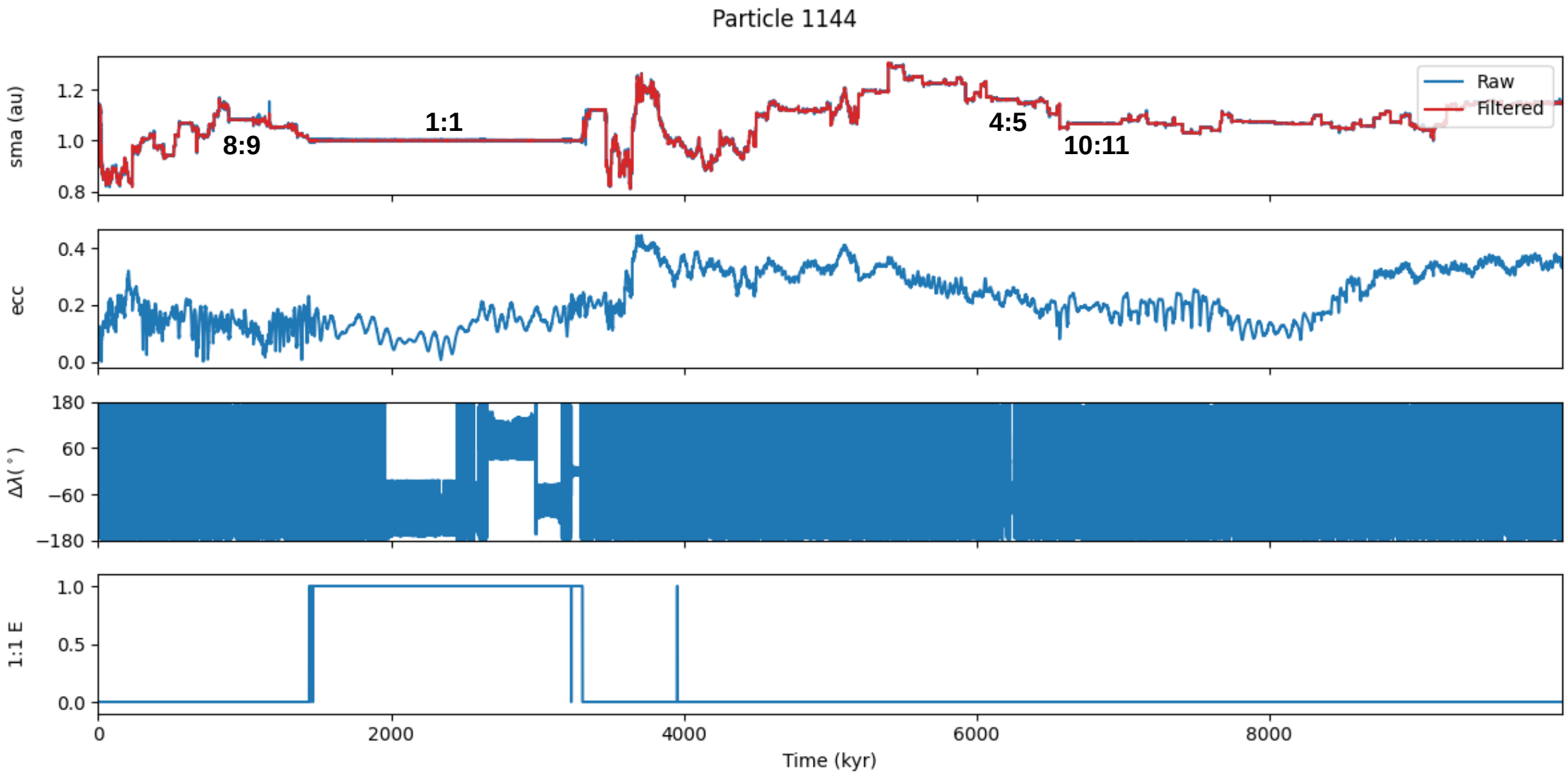
- Integrator: MERCURIUS (hybrid)
- Termination: Collisions with planets, $q < 0.01$ AU or $Q > 6.0$ AU

Phase	Bodies	Time
Geocentric	Sun, Earth, Moon	100 yr
Heliocentric	Sun, 8 planets	10 Myr

Credit: [1] Hanno Rein et al. MNRAS (2019)

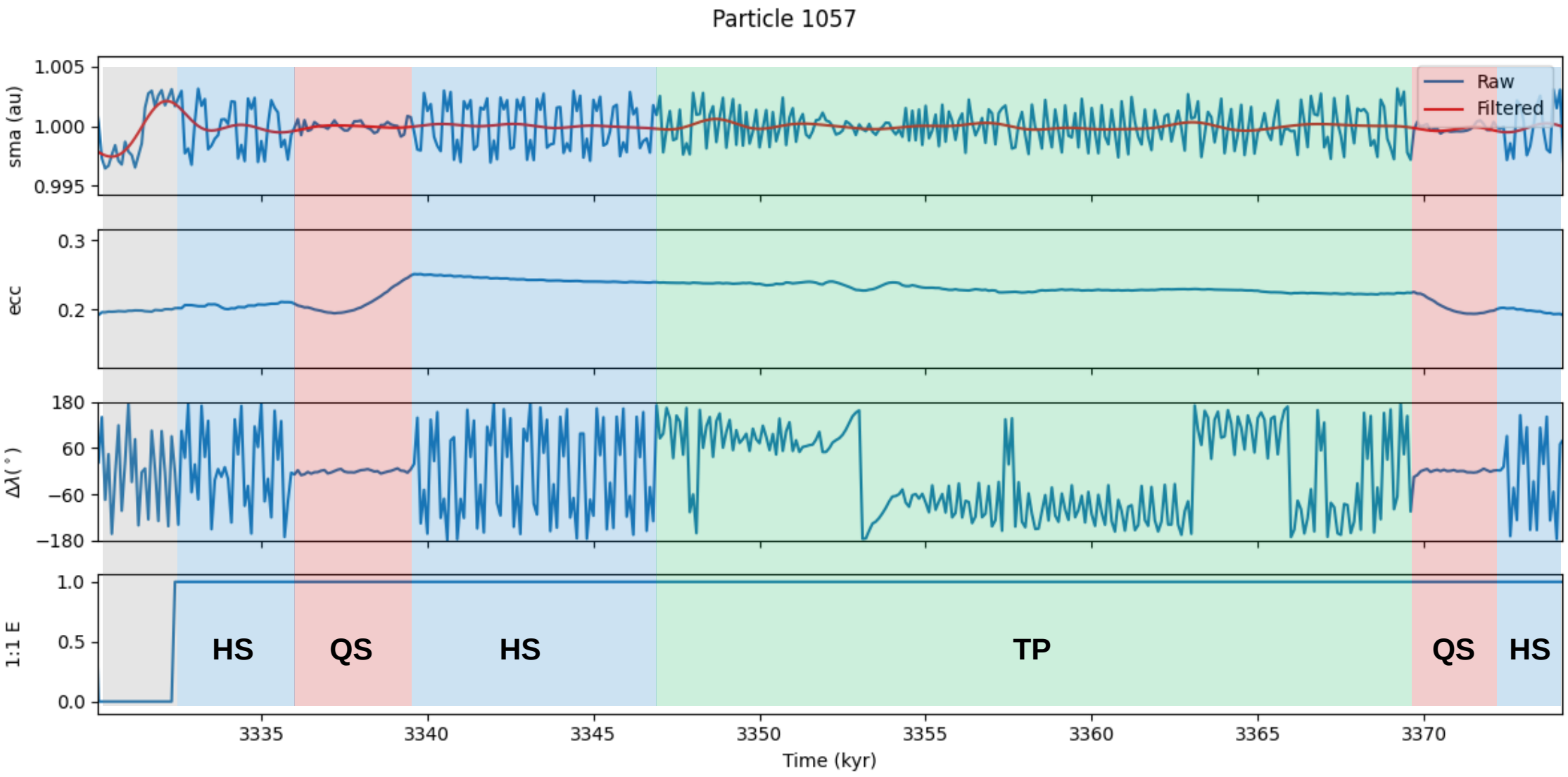
Fragments Dynamical Evolution

Earth Resonances



Fragments Dynamical Evolution

Earth 1:1 Resonance



Fragments Dynamical Evolution

N-body Simulation Results

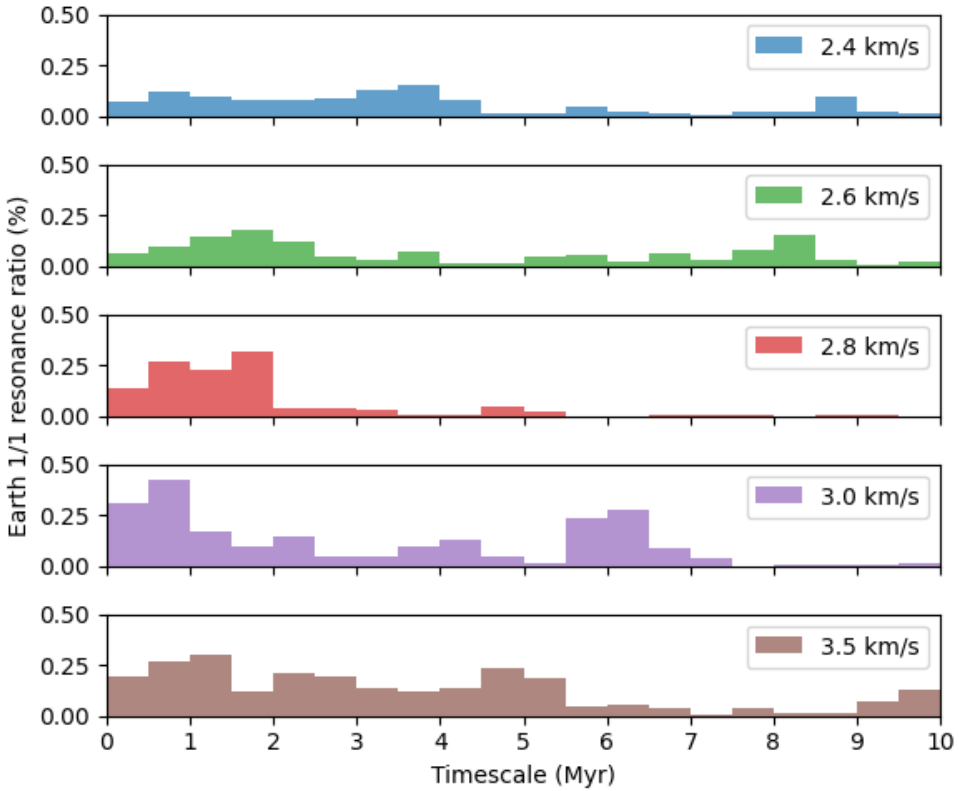
Launch Velocity	2.4 km/s		2.6 km/s		2.8 km/s		3.0 km/s		3.5 km/s	
Phase	Geo	Helio	Geo	Helio	Geo	Helio	Geo	Helio	Geo	Helio
Initial particles	2000	1722	1000	663	1000	628	1000	822	1000	998
Survival particles	1722	217	663	100	628	118	822	183	998	222
Collision with										
Earth	178	1310	329	434	370	380	166	467	0	478
Moon	100	-	8	-	2	-	12	-	2	-
Venus	-	145	-	110	-	106	-	143	-	251
Mercury	-	7	-	4	-	0	-	0	-	7
Mars	-	5	-	1	-	0	-	0	-	1
Q>6.0 au	-	8	-	4	-	8	-	6	-	11
q<0.01 au	-	30	-	10	-	16	-	23	-	28
1:1 resonant ratio %										
0~10 Myr	0.057993		0.063049		0.058093		0.109429		0.126459	

Resonance ratio = Total resonance time / (Initial particles number * Time interval)

Fragments Dynamical Evolution

Earth Co-orbits from GB

3.7~8.0 Earth co-orbital fragments over 10 Myr

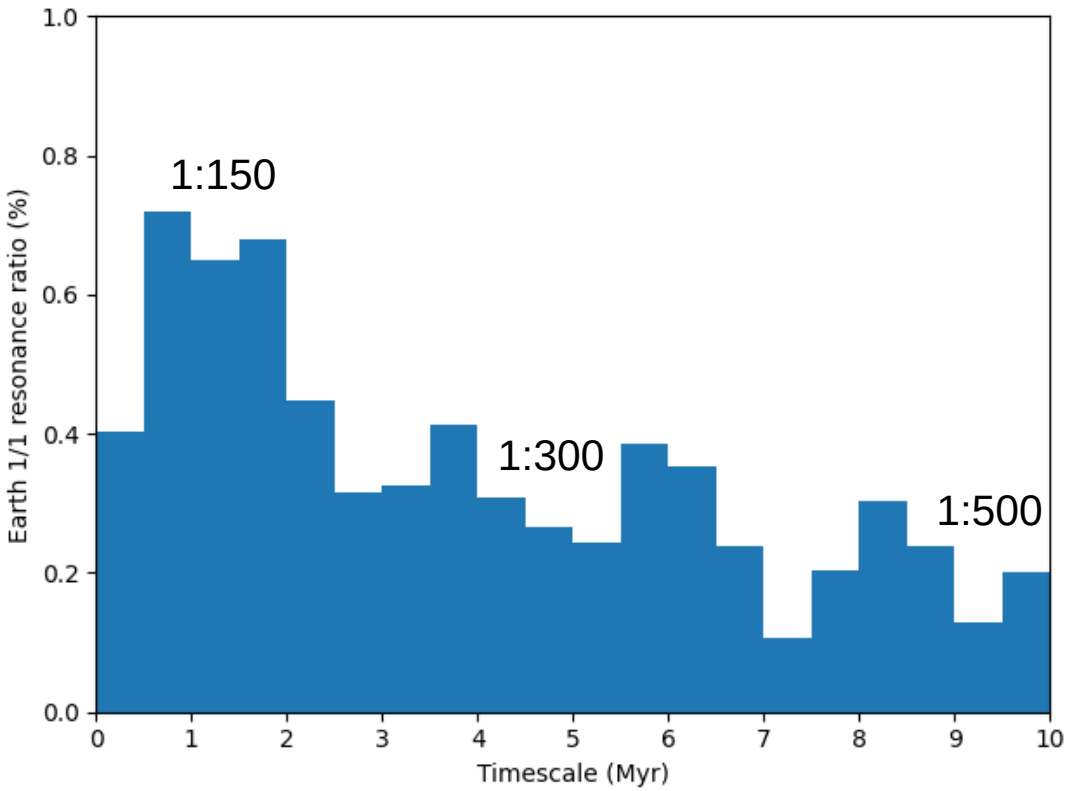


Earth 1:1 resonance ratio

Ratio = Total resonance time / (**Initial** particles number * Time interval)

40-m NEAs: discovered number is ~1% of estimated

1 Earth co-orbit indicates ~300 NEAs from GB



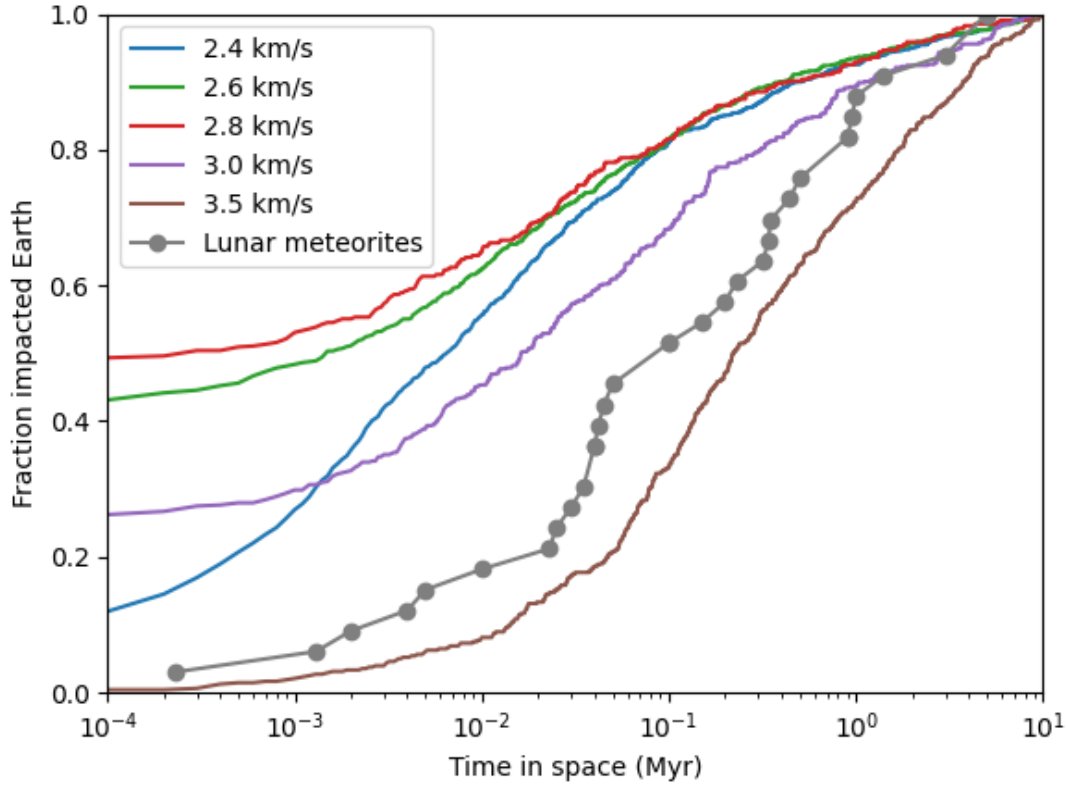
Real-time Earth co-orbits ratio

Ratio = Total resonance time / (**Real-time** particles number * Time interval)

Fragments Dynamical Evolution

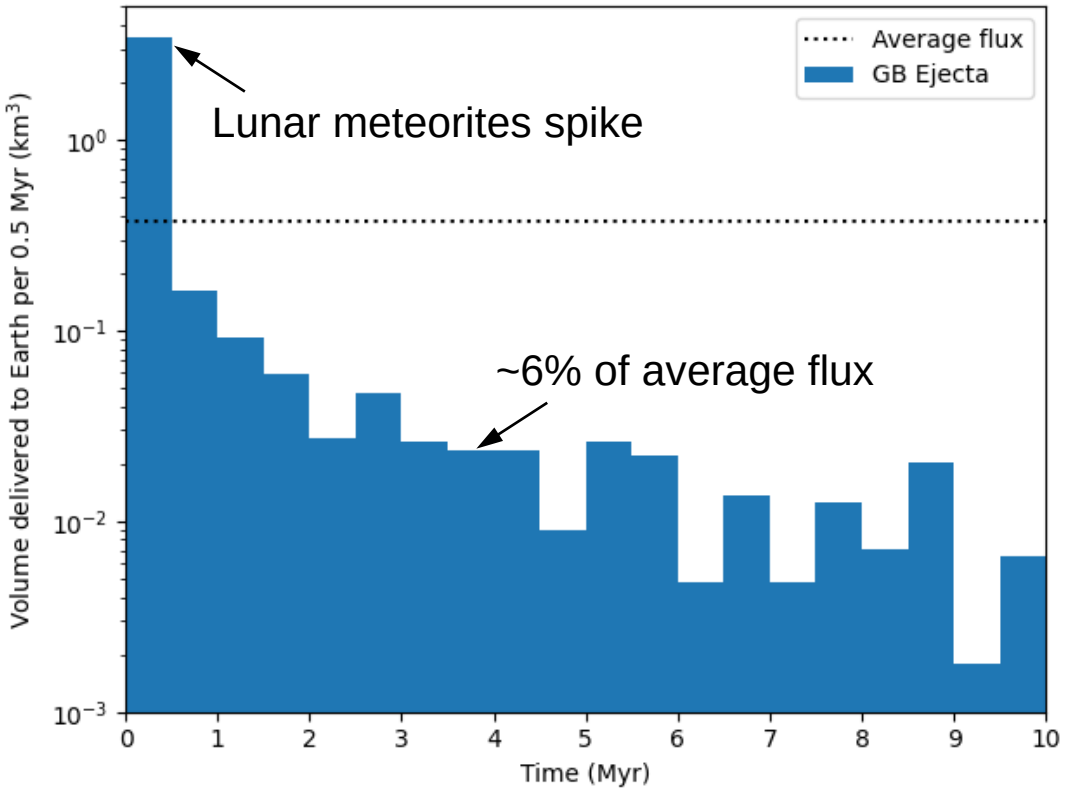
Lunar Meteorites from GB

Lunar meteorites CRE age ~ 1 Myr



Earth collisions fraction and lunar meteorites counting[1]

Lunar meteorites terrestrial age <0.6 Myr[1]



Volume delivered to Earth from GB

Average flux = (Total craters volume[2] per 0.5 Myr / GB volume) * GB ejecta volume * Delivery efficiency

Credit: [1] J. Fritz, Icarus (2012); [2] G. Neukum et al. Space Science Reviews (2001)

Discussion

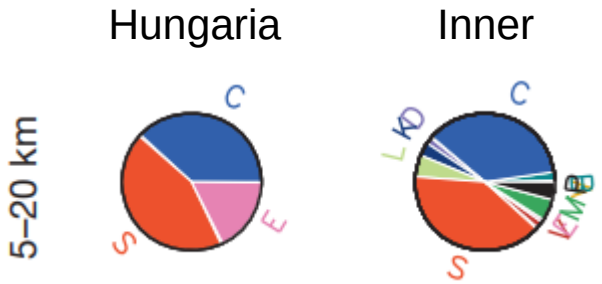
Origin from Main Belt

Potential Source Region from MB

Kamo`oalewa	
Semi-major axis	1.001 AU
Eccentricity	0.103
Inclination	7.792 deg
Absolute magnitude	24.3
Source region	Probability[2]
v6	74±5%
Hungarias	21±5%
3:1 J	5±1%
Estimated ~77 similar NEAs[2]	
Semi-major axis	0.99~1.01 AU
Eccentricity	0~0.3
Inclination	0~15 deg
Absolute magnitude	< 24.3

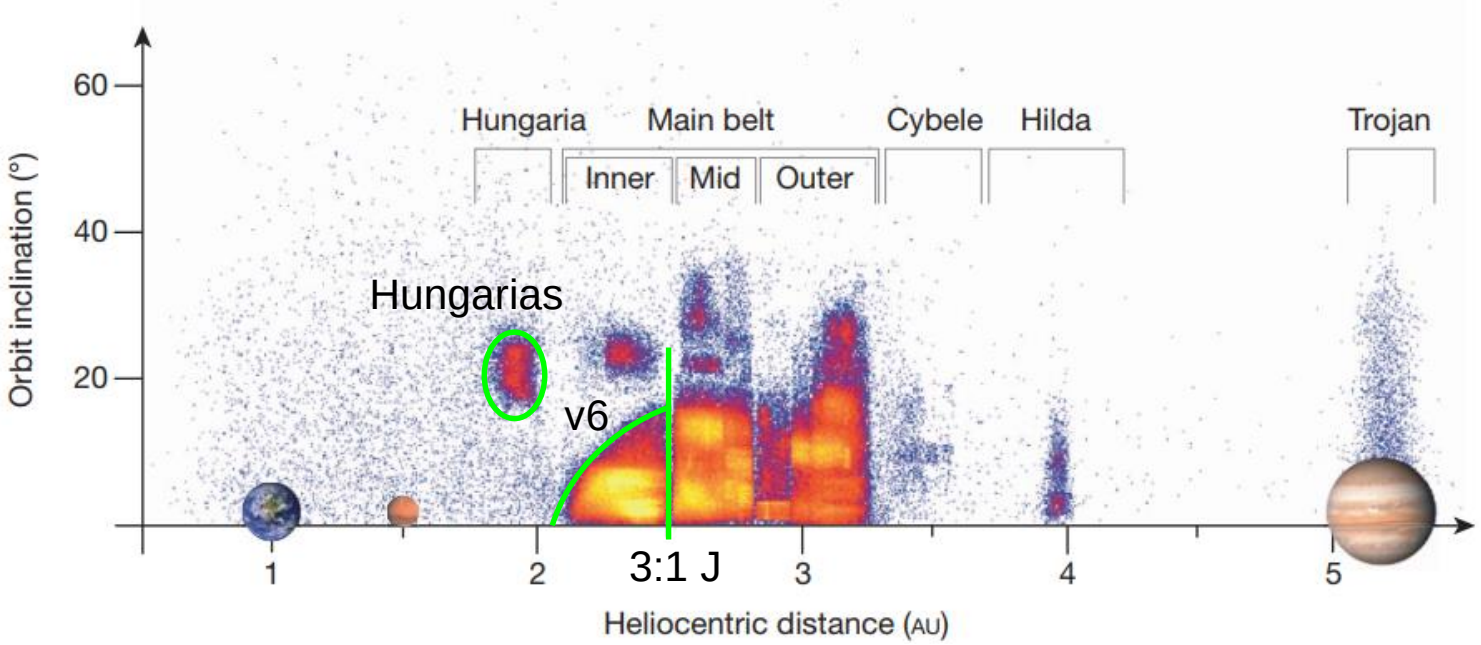
GB ejecta in Earth co-orbits: 5~10% (3.7~8.0 to 77)

GB ejecta in NEAs (H<24.3) : 0.25~0.56% (630~1400 to 250k)



Most similar NEAs from MB are S, C or E-type[1]

✘ Reflectance spectra



Credit: [1] F. E. DeMeo et al. Nature (2014); [2] Mikael Granvik et al. Icarus (2018)

Implications for Structure & Surface

Predictions of Kamo`oalewa

(Spalled & lightly shocked)

Intact & fast-rotating fragment from GB crater

Earth flyby

YORP effect

Thermal fatigue

Meteoroid impacts

~4 Myr

✘ **Rubble-pile structure**

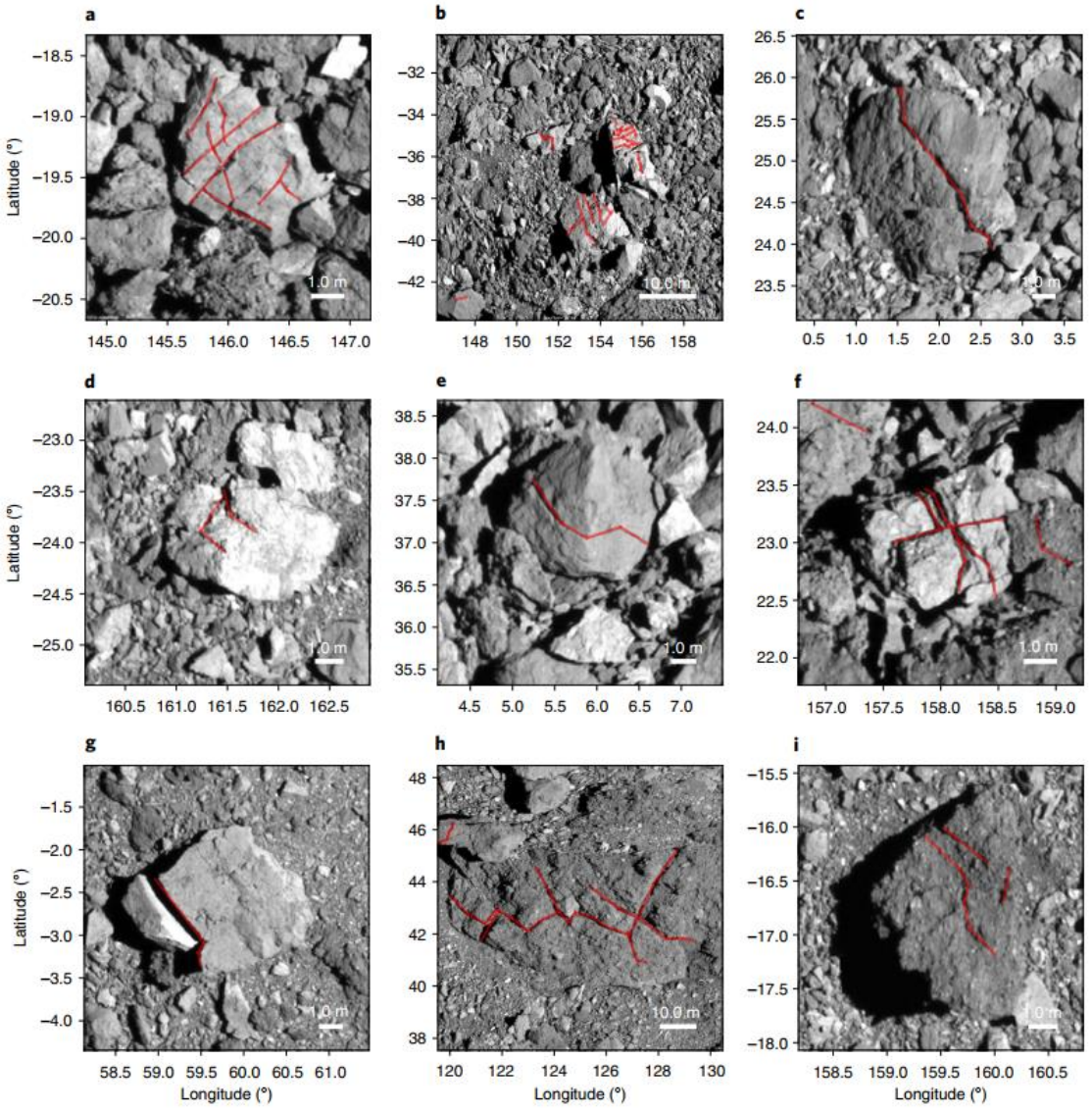
Rock asteroid with fractures and shallow regolith & fast-rotating

Formation timescale

10~100 kyr[1]

Required cohesion

~0.2 Pa[2]



Bennu surface[1]

Credit: [1] Marco Delbo et al. Nature Geoscience (2022); [2] Xiangyu Li et al. Icarus (2021)

Summary

Exploring Lunar Ejecta Origin of Asteroid Kamo`oalewa

Lunar ejecta origin

Kamo`oalewa

- Earth co-orbital
- Small sized
- Fast-rotating

Implications

✘ Rubble-pile structure

☐ Fast-rotating rock asteroid

☐ Fractures and shallow regolith

☐ 3.7~8.0 Earth co-orbits from GB

?

Impact simulation

- 2-km asteroid impact at 18 km/s, 45 deg
- Largest escaped fragment ~71 m
- ~6,300 escaped fragments in 40~60 m

Reflectance spectra

✘ Asteroid taxonomies

✔ Apollo 14 sample

✔ Luna 24 sample

✔ Lunar meteorite

Main belt migration

- Source region: Inner MB, Hungaria
- Estimated ~77 potential Earth co-orbits

Constraints

- Crater size > 10 km
- Crater age ~ 10 Myr

Giordano Bruno crater

- 22 km diameter & ~4 Myr age
- Most possible source crater

N-body simulation

- Launch particles from GB crater
- Most terminate in collisions
- 10~22% survive in 10 Myr
- 0.06~0.13% 1:1 E in 10 Myr